
TP1 BDA

ING2 INFORMATIQUE
Sup Galilée

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1 Exercice 1

1.1 Diagramme Entité-Association

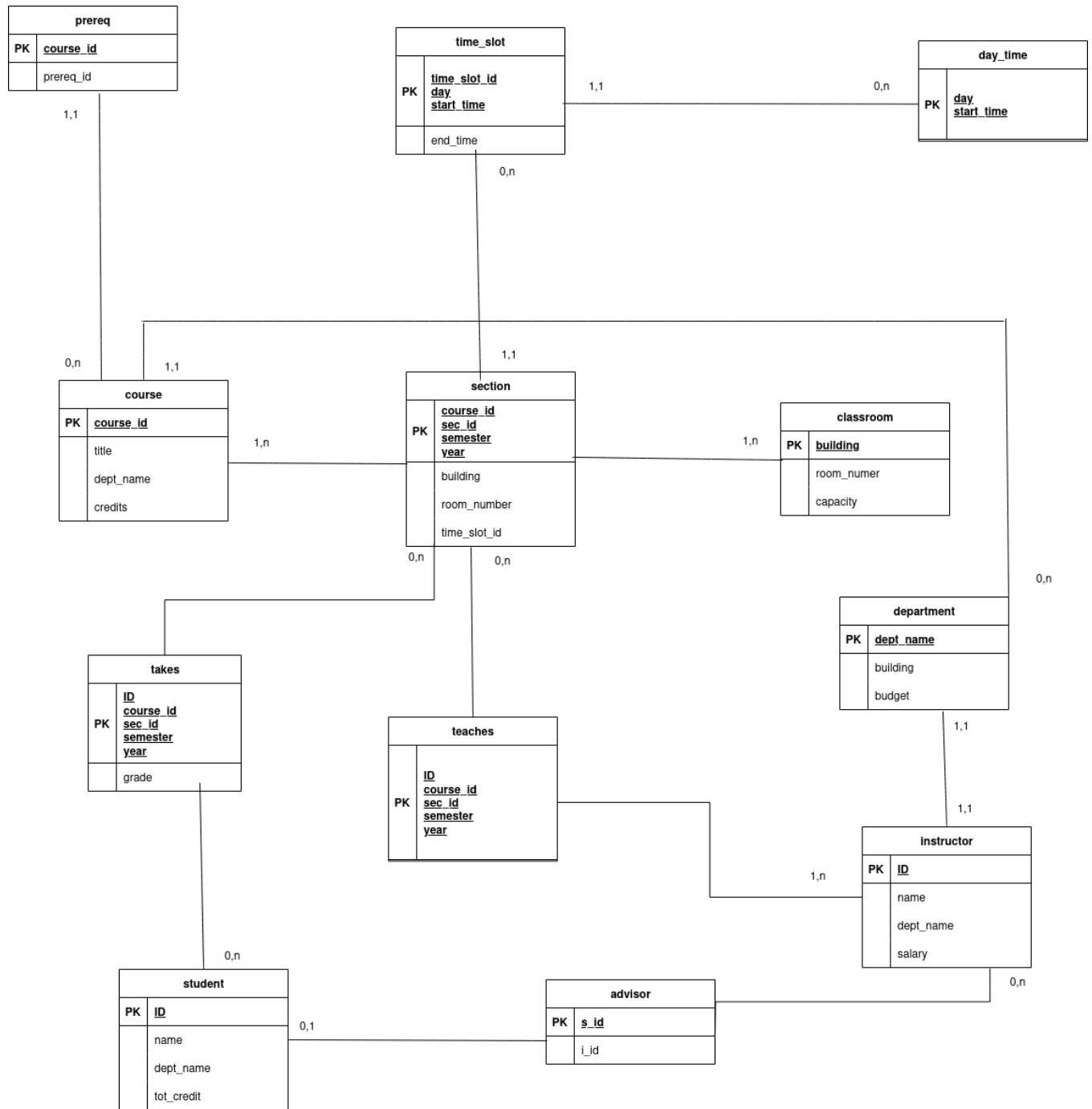


Figure 1: Diagramme Entité-Association

2 Exercice 2

1.

```
SELECT * FROM section NATURAL JOIN course;
```

2.

```
SELECT * FROM course;
```

3.

```
SELECT title, dept_name FROM course;
```

4.

```
SELECT dept_name, budget FROM department;
```

5.

```
SELECT name, dept_name FROM teacher;
```

6.

```
SELECT name FROM teacher WHERE salary > 65000;
```

7.

```
SELECT name FROM teacher WHERE salary BETWEEN 55000 AND 85000;
```

8.

```
SELECT DISTINCT dept_name FROM teacher;
```

9.

```
SELECT name FROM teacher WHERE salary > 65000 AND dept_name = 'Comp. Sci.';
```

10.

```
SELECT * FROM section WHERE semester = 'Spring' AND year = 2010;
```

11.

```
SELECT title FROM course WHERE credits > 3 AND dept_name = 'Comp. Sci.';
```

12.

```
SELECT name, dept_name, building FROM teacher NATURAL JOIN department;
```

13.

```
SELECT DISTINCT student.*  
FROM takes, student, course  
WHERE takes.ID = student.ID  
      AND takes.course_id = course.course_id  
      AND course.dept_name = 'Comp. Sci.';
```

14.

```
SELECT DISTINCT student.name  
FROM takes, student, course, teacher, teaches  
WHERE takes.ID = student.ID  
      AND takes.course_id = course.course_id  
      AND teacher.name = 'Einstein'  
      AND teaches.course_id = takes.course_id  
      AND teaches.ID = teacher.ID;
```

15.

```
SELECT DISTINCT course.course_id, teacher.name  
FROM (teaches NATURAL JOIN course) NATURAL JOIN teacher;
```

16.

```
SELECT DISTINCT course_id, name  
FROM (teaches NATURAL JOIN course) NATURAL JOIN teacher;
```

17.

```
SELECT course_id, count(ID)
FROM ((takes NATURAL JOIN student) NATURAL JOIN student)
WHERE semester = 'Spring' AND year = 2010
GROUP BY course_id;
```

18.

```
SELECT dept_name, max(salary)
FROM (department NATURAL JOIN teacher)
GROUP BY dept_name;
```

19.

```
SELECT course_id, count(ID)
FROM (takes NATURAL JOIN student)
GROUP BY course_id
ORDER BY count(ID) DESC;
```

20.

```
SELECT building, count(course_id)
FROM section
WHERE (semester = 'Autumn' AND year = 2010)
      OR (semester = 'Spring' AND year = 2010)
GROUP BY building;
```

21.

```
SELECT dept_name, count(course_id)
FROM Department NATURAL JOIN course
GROUP BY dept_name;
```

22.

```
SELECT dept_name, count(course_id)
FROM (((section NATURAL JOIN course) NATURAL JOIN teaches) NATURAL JOIN teacher) NATURAL JOIN department
GROUP BY dept_name;
```

23.

```
SELECT DISTINCT title, name
FROM ((section NATURAL JOIN teaches) NATURAL JOIN teacher) NATURAL JOIN course;
```

24.

```
SELECT semester, count(course_id)
FROM section
GROUP BY semester;
```

25.

```
SELECT student.name, sum(credits)
FROM (student NATURAL JOIN takes), course
WHERE course.course_id = takes.course_id
      AND student.dept_name <> course.dept_name
GROUP BY id, name;
```